

Thành viên nhóm 4:

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Einstein's Riddle (The Zebra Puzzle)

1. The Situation

Có 5 ngôi nhà với 5 màu sắc khác nhau. Trong mỗi nhà có một người với quốc tịch khác nhau. 5 chủ nhà này uống các loại đồ uống khác nhau, hút các loại thuốc lá khác nhau và nuôi các loại thú cưng khác nhau. Không có ai có cùng thú cưng, hút cùng loại thuốc hay uống cùng một loại đồ uống.

2. The 15 Clues

1. The Brit lives in the Red house. (Người Anh ở nhà đỏ).
2. The Spaniard owns the Dog. (Người Tây Ban Nha nuôi chó).
3. Coffee is drunk in the Green house. (Cà phê được uống ở nhà xanh lá).
4. The Ukrainian drinks Tea. (Người Ukraine uống trà).
5. The Green house is immediately to the right of the Ivory house. (Nhà xanh lá nằm ngay bên phải nhà màu ngà).
6. The Old Gold smoker owns Snails. (Người hút thuốc Old Gold nuôi ốc sên).
7. Kools are smoked in the Yellow house. (Thuốc Kools được hút ở nhà vàng).
8. Milk is drunk in the middle house. (Sữa được uống ở nhà chính giữa- nhà số 3).
9. The Norwegian lives in the first house. (Người Na Uy ở nhà đầu tiên- nhà số 1).
10. The man who smokes Chesterfields lives in the house next to the man with the Fox. (Người hút Chesterfields ở cạnh nhà người nuôi cáo).
11. Kools are smoked in the house next to the house where the Horse is kept. (Thuốc Kools được hút ở cạnh nhà nuôi ngựa).

12. The Lucky Strike smoker drinks Orange Juice. (Người hút Lucky Strike uống nước cam).

13. The Japanese smokes Parliaments. (Người Nhật hút thuốc Parliaments).

14. The Norwegian lives next to the Blue house. (Người Na Uy ở cạnh nhà màu xanh lam).

15. The man who smokes Chesterfields has a neighbor who drinks Water. (Người hút Chesterfields có hàng xóm uống nước).

3. Question: Who owns the Zebra? Who drinks Water?

Requirements:

- AC-3
- Backtracking

Bài làm

1. Code:

- Link git: <https://github.com/DannyTuanAnh/The-Zebra-Puzzle>

2. Giải tay:

- Ta có 5 biến cho mỗi thuộc tính (tổng cộng 25 biến):

+ **Nationality:** Nor, Ukr, Bri, Spa, Jap

+ **Color:** Red, Blu, Gre, Ivo, Yel

+ **Drink:** Mil, Tea, Cof, Jui, Wat

+ **Tobacco:** Koo, Che, OG, LS, Par

+ **Pet:** Fox, Hor, Sna, Dog, Zeb

- Với mọi biến, miền giá trị ban đầu là $D = \{1, 2, 3, 4, 5\}$

- *Ràng buộc:*

1. Drink(3) = Mil

2. Nationality(1) = Nor
3. Nationality(i) = Bri $\Leftrightarrow$ Color(i) = Red
4. Nationality(i) = Spa $\Leftrightarrow$ Pet(i) = Dog
5. Color(i) = Gre $\Leftrightarrow$ Drink(i) = Cof
6. Nationality(i) = Ukr $\Leftrightarrow$ Drink(i) = Tea
7. Tobacco(i) = OG $\Leftrightarrow$ Pet(i) = Sna
8. Tobacco(i) = Koo $\Leftrightarrow$ Color(i) = Yel
9. Tobacco(i) = LS $\Leftrightarrow$ Drink(i) = Jui
10. Nationality(i) = Jap $\Leftrightarrow$ Tobacco(i) = Par
11. Color(i) = Ivo $\Leftrightarrow$ Color(i+1) = Gre (với $i \in \{1,2,3,4\}$)
12. Tobacco(i) = Che $\Leftrightarrow$ [Pet(i-1) = Fox] $\vee$ [Pet(i+1) = Fox]
13. Tobacco(i) = Koo $\Leftrightarrow$ [Pet(i-1) = Hor] $\vee$ [Pet(i+1) = Hor]
14. Nationality(i) = Nor $\Leftrightarrow$ [Color(i-1) = Blu] $\vee$ [Color(i+1) = Blu]
15. Tobacco(i) = Che $\Leftrightarrow$ [Drink(i-1) = Wat] $\vee$ [Drink(i+1) = Wat]
16. All-Different: Tất cả 5 giá trị trong cùng một hạng mục (ví dụ 5 quốc tịch) phải khác nhau:

D(X)	
Nor{1, 2, 3, 4, 5}, Ukr{2, 3, 4, 5}, Bri{2, 3, 4, 5}, Spa{2, 3, 4, 5}, Jap{2, 3, 4, 5} Blu{1, 2, 3, 4, 5}, Red{1, 2, 3, 4, 5}, Gre{1, 2, 3, 4, 5}, Ivo{1, 2, 3, 4, 5}, Yel{1, 2, 3, 4, 5}, Mil{3}, Tea{1, 2, 4, 5}, Cof{1, 2, 4, 5}, Jui{1, 2, 4, 5}, Wat{1, 2, 4, 5},	1. Drink(3) = Mil 16. All-Different

Koo{1, 2, 3, 4, 5}, Che{1, 2, 3, 4, 5}, OG{1, 2, 3, 4, 5}, LS{1, 2, 3, 4, 5}, Par{1, 2, 3, 4, 5}, Fox{1, 2, 3, 4, 5}, Hor{1, 2, 3, 4, 5}, Sna{1, 2, 3, 4, 5}, Dog{1, 2, 3, 4, 5}, Zeb{1, 2, 3, 4, 5}	
Nor{1}, Ukr{2, 3, 4, 5}, Bri{2, 3, 4, 5}, Spa{2, 3, 4, 5}, Jap{2, 3, 4, 5} Blu{1, 2, 3, 4, 5}, Red{1, 2, 3, 4, 5}, Gre{1, 2, 3, 4, 5}, Ivo{1, 2, 3, 4, 5}, Yel{1, 2, 3, 4, 5}, Mil{3}, Tea{1, 2, 4, 5}, Cof{1, 2, 4, 5}, Jui{1, 2, 4, 5}, Wat{1, 2, 4, 5}, Koo{1, 2, 3, 4, 5}, Che{1, 2, 3, 4, 5}, OG{1, 2, 3, 4, 5}, LS{1, 2, 3, 4, 5}, Par{1, 2, 3, 4, 5}, Fox{1, 2, 3, 4, 5}, Hor{1, 2, 3, 4, 5}, Sna{1, 2, 3, 4, 5}, Dog{1, 2, 3, 4, 5}, Zeb{1, 2, 3, 4, 5}	2. Nationality(1) = Nor 16. All-Different
Nor{1}, Ukr{2, 3, 4, 5}, Bri{2, 3, 4, 5}, Spa{2, 3, 4, 5}, Jap{2, 3, 4, 5} Blu{1, 2, 3, 4, 5}, Red{2, 3, 4, 5}, Gre{1, 2, 3, 4, 5}, Ivo{1, 2, 3, 4, 5}, Yel{1, 2, 3, 4, 5}, Mil{3}, Tea{1, 2, 4, 5}, Cof{1, 2, 4, 5}, Jui{1, 2, 4, 5}, Wat{1, 2, 4, 5}, Koo{1, 2, 3, 4, 5}, Che{1, 2, 3, 4, 5}, OG{1, 2, 3, 4, 5}, LS{1, 2, 3, 4, 5}, Par{1, 2, 3, 4, 5}, Fox{1, 2, 3, 4, 5}, Hor{1, 2, 3, 4, 5}, Sna{1, 2, 3, 4, 5}, Dog{1, 2, 3, 4, 5}, Zeb{1, 2, 3, 4, 5}	3. Nationality(i) = Bri ⇔ Color(i) = Red
Nor{1}, Ukr{2, 3, 4, 5}, Bri{2, 3, 4, 5}, Spa{2, 3, 4, 5}, Jap{2, 3, 4, 5}	4. Nationality(i) = Spa ⇔ Pet(i) = Dog

Blu{1, 2, 3, 4, 5}, Red{2, 3, 4, 5}, Gre{1, 2, 3, 4, 5}, Ivo{1, 2, 3, 4, 5}, Yel{1, 2, 3, 4, 5}, Mil{3}, Tea{1, 2, 4, 5}, Cof{1, 2, 4, 5}, Jui{1, 2, 4, 5}, Wat{1, 2, 4, 5}, Koo{1, 2, 3, 4, 5}, Che{1, 2, 3, 4, 5}, OG{1, 2, 3, 4, 5}, LS{1, 2, 3, 4, 5}, Par{1, 2, 3, 4, 5}, Fox{1, 2, 3, 4, 5}, Hor{1, 2, 3, 4, 5}, Sna{1, 2, 3, 4, 5}, Dog{2, 3, 4, 5}, Zeb{1, 2, 3, 4, 5}	
Nor{1}, Ukr{2, 3, 4, 5}, Bri{2, 3, 4, 5}, Spa{2, 3, 4, 5}, Jap{2, 3, 4, 5} Blu{1, 2, 3, 4, 5}, Red{2, 3, 4, 5}, Gre{1, 2, 4, 5}, Ivo{1, 2, 3, 4, 5}, Yel{1, 2, 3, 4, 5}, Mil{3}, Tea{1, 2, 4, 5}, Cof{1, 2, 4, 5}, Jui{1, 2, 4, 5}, Wat{1, 2, 4, 5}, Koo{1, 2, 3, 4, 5}, Che{1, 2, 3, 4, 5}, OG{1, 2, 3, 4, 5}, LS{1, 2, 3, 4, 5}, Par{1, 2, 3, 4, 5}, Fox{1, 2, 3, 4, 5}, Hor{1, 2, 3, 4, 5}, Sna{1, 2, 3, 4, 5}, Dog{2, 3, 4, 5}, Zeb{1, 2, 3, 4, 5}	5. Color(i) = Gre ⇔ Drink(i) = Cof
Nor{1}, Ukr{2, 4, 5}, Bri{2, 3, 4, 5}, Spa{2, 3, 4, 5}, Jap{2, 3, 4, 5} Blu{1, 2, 3, 4, 5}, Red{2, 3, 4, 5}, Gre{1, 2, 4, 5}, Ivo{1, 2, 3, 4, 5}, Yel{1, 2, 3, 4, 5}, Mil{3}, Tea{2, 4, 5}, Cof{1, 2, 4, 5}, Jui{1, 2, 4, 5}, Wat{1, 2, 4, 5}, Koo{1, 2, 3, 4, 5}, Che{1, 2, 3, 4, 5}, OG{1, 2, 3, 4, 5}, LS{1, 2, 3, 4, 5}, Par{1, 2, 3, 4, 5},	6. Nationality(i) = Ukr ⇔ Drink(i) = Tea

Fox{1, 2, 3, 4, 5}, Hor{1, 2, 3, 4, 5}, Sna{1, 2, 3, 4, 5}, Dog{2, 3, 4, 5}, Zeb{1, 2, 3, 4, 5}	
Nor{1}, Ukr{2, 4, 5}, Bri{2, 3, 4, 5}, Spa{2, 3, 4, 5}, Jap{2, 3, 4, 5} Blu{1, 2, 3, 4, 5}, Red{2, 3, 4, 5}, Gre{1, 2, 4, 5}, Ivo{1, 2, 3, 4, 5}, Yel{1, 2, 3, 4, 5}, Mil{3}, Tea{2, 4, 5}, Cof{1, 2, 4, 5}, Jui{1, 2, 4, 5}, Wat{1, 2, 4, 5}, Koo{1, 2, 3, 4, 5}, Che{1, 2, 3, 4, 5}, OG{1, 2, 3, 4, 5}, LS{1, 2, 3, 4, 5}, Par{1, 2, 3, 4, 5}, Fox{1, 2, 3, 4, 5}, Hor{1, 2, 3, 4, 5}, Sna{1, 2, 3, 4, 5}, Dog{2, 3, 4, 5}, Zeb{1, 2, 3, 4, 5}	7. Tobacco(i) = OG ⇔ Pet(i) = Sna
Nor{1}, Ukr{2, 4, 5}, Bri{2, 3, 4, 5}, Spa{2, 3, 4, 5}, Jap{2, 3, 4, 5} Blu{1, 2, 3, 4, 5}, Red{2, 3, 4, 5}, Gre{1, 2, 4, 5}, Ivo{1, 2, 3, 4, 5}, Yel{1, 2, 3, 4, 5}, Mil{3}, Tea{2, 4, 5}, Cof{1, 2, 4, 5}, Jui{1, 2, 4, 5}, Wat{1, 2, 4, 5}, Koo{1, 2, 3, 4, 5}, Che{1, 2, 3, 4, 5}, OG{1, 2, 3, 4, 5}, LS{1, 2, 3, 4, 5}, Par{1, 2, 3, 4, 5}, Fox{1, 2, 3, 4, 5}, Hor{1, 2, 3, 4, 5}, Sna{1, 2, 3, 4, 5}, Dog{2, 3, 4, 5}, Zeb{1, 2, 3, 4, 5}	8. Tobacco(i) = Koo ⇔ Color(i) = Yel
Nor{1}, Ukr{2, 4, 5}, Bri{2, 3, 4, 5}, Spa{2, 3, 4, 5}, Jap{2, 3, 4, 5} Blu{1, 2, 3, 4, 5}, Red{2, 3, 4, 5}, Gre{1, 2, 4, 5}, Ivo{1, 2, 3, 4, 5}, Yel{1, 2, 3, 4, 5}, Mil{3}, Tea{2, 4, 5}, Cof{1, 2, 4, 5}, Jui{1, 2, 4, 5}, Wat{1, 2, 4, 5},	9. Tobacco(i) = LS ⇔ Drink(i) = Jui

Koo{1, 2, 3, 4, 5}, Che{1, 2, 3, 4, 5}, OG{1, 2, 3, 4, 5}, LS{1, 2, 4, 5}, Par{1, 2, 3, 4, 5}, Fox{1, 2, 3, 4, 5}, Hor{1, 2, 3, 4, 5}, Sna{1, 2, 3, 4, 5}, Dog{2, 3, 4, 5}, Zeb{1, 2, 3, 4, 5}	
Nor{1}, Ukr{2, 4, 5}, Bri{2, 3, 4, 5}, Spa{2, 3, 4, 5}, Jap{2, 3, 4, 5} Blu{1, 2, 3, 4, 5}, Red{2, 3, 4, 5}, Gre{1, 2, 4, 5}, Ivo{1, 2, 3, 4, 5}, Yel{1, 2, 3, 4, 5}, Mil{3}, Tea{2, 4, 5}, Cof{1, 2, 4, 5}, Jui{1, 2, 4, 5}, Wat{1, 2, 4, 5}, Koo{1, 2, 3, 4, 5}, Che{1, 2, 3, 4, 5}, OG{1, 2, 3, 4, 5}, LS{1, 2, 4, 5}, Par{2, 3, 4, 5}, Fox{1, 2, 3, 4, 5}, Hor{1, 2, 3, 4, 5}, Sna{1, 2, 3, 4, 5}, Dog{2, 3, 4, 5}, Zeb{1, 2, 3, 4, 5}	10. Nationality(i) = Jap $\Leftrightarrow$ Tobacco(i) = Par
Nor{1}, Ukr{2, 4, 5}, Bri{2, 3, 4, 5}, Spa{2, 3, 4, 5}, Jap{2, 3, 4, 5} Blu{1, 2, 3, 4, 5}, Red{2, 3, 4, 5}, Gre{2, 4, 5}, Ivo{1, 3, 4}, Yel{1, 2, 3, 4, 5}, Mil{3}, Tea{2, 4, 5}, Cof{1, 2, 4, 5}, Jui{1, 2, 4, 5}, Wat{1, 2, 4, 5}, Koo{1, 2, 3, 4, 5}, Che{1, 2, 3, 4, 5}, OG{1, 2, 3, 4, 5}, LS{1, 2, 4, 5}, Par{2, 3, 4, 5}, Fox{1, 2, 3, 4, 5}, Hor{1, 2, 3, 4, 5}, Sna{1, 2, 3, 4, 5}, Dog{2, 3, 4, 5}, Zeb{1, 2, 3, 4, 5}	11. Color(i) = Ivo $\Leftrightarrow$ Color(i+1) = Gre (với i $\in$ {1,2,3,4})
Nor{1}, Ukr{2, 4, 5}, Bri{2, 3, 4, 5}, Spa{2, 3, 4, 5}, Jap{2, 3, 4, 5} Blu{1, 2, 3, 4, 5}, Red{2, 3, 4, 5}, Gre{2, 4, 5}, Ivo{1, 3, 4}, Yel{1, 2, 3, 4, 5},	12. Tobacco(i) = Che $\Leftrightarrow$ [Pet(i-1) = Fox] $\vee$ [Pet(i+1) = Fox]

Mil{3}, Tea{2, 4, 5}, Cof{1, 2, 4, 5}, Jui{1, 2, 4, 5}, Wat{1, 2, 4, 5}, Koo{1, 2, 3, 4, 5}, Che{1, 2, 3, 4, 5}, OG{1, 2, 3, 4, 5}, LS{1, 2, 4, 5}, Par{2, 3, 4, 5}, Fox{1, 2, 3, 4, 5}, Hor{1, 2, 3, 4, 5}, Sna{1, 2, 3, 4, 5}, Dog{2, 3, 4, 5}, Zeb{1, 2, 3, 4, 5}	
Nor{1}, Ukr{2, 4, 5}, Bri{2, 3, 4, 5}, Spa{2, 3, 4, 5}, Jap{2, 3, 4, 5} Blu{1, 2, 3, 4, 5}, Red{2, 3, 4, 5}, Gre{2, 4, 5}, Ivo{1, 3, 4}, Yel{1, 2, 3, 4, 5}, Mil{3}, Tea{2, 4, 5}, Cof{1, 2, 4, 5}, Jui{1, 2, 4, 5}, Wat{1, 2, 4, 5}, Koo{1, 2, 3, 4, 5}, Che{1, 2, 3, 4, 5}, OG{1, 2, 3, 4, 5}, LS{1, 2, 4, 5}, Par{2, 3, 4, 5}, Fox{1, 2, 3, 4, 5}, Hor{1, 2, 3, 4, 5}, Sna{1, 2, 3, 4, 5}, Dog{2, 3, 4, 5}, Zeb{1, 2, 3, 4, 5}	13. Tobacco(i) = Koo $\Leftrightarrow$ $[\text{Pet}(i-1) = \text{Hor}] \vee$ $[\text{Pet}(i+1) = \text{Hor}]$
Nor{1}, Ukr{2, 4, 5}, Bri{2, 3, 4, 5}, Spa{2, 3, 4, 5}, Jap{2, 3, 4, 5} Blu{2}, Red{3, 4, 5}, Gre{4, 5}, Ivo{1, 3, 4}, Yel{1, 3, 4, 5}, Mil{3}, Tea{2, 4, 5}, Cof{1, 2, 4, 5}, Jui{1, 2, 4, 5}, Wat{1, 2, 4, 5}, Koo{1, 2, 3, 4, 5}, Che{1, 2, 3, 4, 5}, OG{1, 2, 3, 4, 5}, LS{1, 2, 4, 5}, Par{2, 3, 4, 5}, Fox{1, 2, 3, 4, 5}, Hor{1, 2, 3, 4, 5}, Sna{1, 2, 3, 4, 5}, Dog{2, 3, 4, 5}, Zeb{1, 2, 3, 4, 5}	14. Nationality(i) = Nor $\Leftrightarrow [\text{Color}(i-1) = \text{Blu}] \vee$ $[\text{Color}(i+1) = \text{Blu}]$ 16. All diff

Nor{1}, Ukr{2, 4, 5}, Bri{2, 3, 4, 5}, Spa{2, 3, 4, 5}, Jap{2, 3, 4, 5} Blu{2}, Red{3, 4, 5}, Gre{4, 5}, Ivo{1, 3, 4}, Yel{1, 3, 4, 5}, Mil{3}, Tea{2, 4, 5}, Cof{1, 2, 4, 5}, Jui{1, 2, 4, 5}, Wat{1, 2, 4, 5}, Koo{1, 2, 3, 4, 5}, Che{1, 2, 3, 4, 5}, OG{1, 2, 3, 4, 5}, LS{1, 2, 4, 5}, Par{2, 3, 4, 5}, Fox{1, 2, 3, 4, 5}, Hor{1, 2, 3, 4, 5}, Sna{1, 2, 3, 4, 5}, Dog{2, 3, 4, 5}, Zeb{1, 2, 3, 4, 5}	15. Tobacco(i) = Che $\Leftrightarrow$ [Drink(i-1) = Wat] $\vee$ [Drink(i+1) = Wat]
Nor{1}, Ukr{2, 4, 5}, Bri{3, 4, 5}, Spa{2, 3, 4, 5}, Jap{2, 3, 4, 5} Blu{2}, Red{3, 4, 5}, Gre{4, 5}, Ivo{1, 3, 4}, Yel{1, 3, 4, 5}, Mil{3}, Tea{2, 4, 5}, Cof{1, 2, 4, 5}, Jui{1, 2, 4, 5}, Wat{1, 2, 4, 5}, Koo{1, 2, 3, 4, 5}, Che{1, 2, 3, 4, 5}, OG{1, 2, 3, 4, 5}, LS{1, 2, 4, 5}, Par{2, 3, 4, 5}, Fox{1, 2, 3, 4, 5}, Hor{1, 2, 3, 4, 5}, Sna{1, 2, 3, 4, 5}, Dog{2, 3, 4, 5}, Zeb{1, 2, 3, 4, 5}	3. Nationality(i) = Bri $\Leftrightarrow$ Color(i) = Red
Nor{1}, Ukr{2, 4, 5}, Bri{3, 4, 5}, Spa{2, 3, 4, 5}, Jap{2, 3, 4, 5} Blu{2}, Red{3, 4, 5}, Gre{4, 5}, Ivo{1, 3, 4}, Yel{1, 3, 4, 5}, Mil{3}, Tea{2, 4, 5}, Cof{4, 5}, Jui{1, 2, 4, 5}, Wat{1, 2, 4, 5}, Koo{1, 2, 3, 4, 5}, Che{1, 2, 3, 4, 5}, OG{1, 2, 3, 4, 5}, LS{1, 2, 4, 5}, Par{2, 3, 4, 5},	5. Color(i) = Gre $\Leftrightarrow$ Drink(i) = Cof

Fox{1, 2, 3, 4, 5}, Hor{1, 2, 3, 4, 5}, Sna{1, 2, 3, 4, 5}, Dog{2, 3, 4, 5}, Zeb{1, 2, 3, 4, 5}	
Nor{1}, Ukr{2, 4, 5}, Bri{3, 4, 5}, Spa{2, 3, 4, 5}, Jap{2, 3, 4, 5} Blu{2}, Red{3, 4, 5}, Gre{4, 5}, Ivo{1, 3, 4}, Yel{1, 3, 4, 5}, Mil{3}, Tea{2, 4, 5}, Cof{4, 5}, Jui{1, 2, 4, 5}, Wat{1, 2, 4, 5}, Koo{1, 3, 4, 5}, Che{1, 2, 3, 4, 5}, OG{1, 2, 3, 4, 5}, LS{1, 2, 4, 5}, Par{2, 3, 4, 5}, Fox{1, 2, 3, 4, 5}, Hor{1, 2, 3, 4, 5}, Sna{1, 2, 3, 4, 5}, Dog{2, 3, 4, 5}, Zeb{1, 2, 3, 4, 5}	8. Tobacco(i) = Koo $\Leftrightarrow$ Color(i) = Yel
Nor{1}, Ukr{2, 4, 5}, Bri{3, 4, 5}, Spa{2, 3, 4, 5}, Jap{2, 3, 4, 5} Blu{2}, Red{3, 4, 5}, Gre{4, 5}, Ivo{3, 4}, Yel{1, 3, 4, 5}, Mil{3}, Tea{2, 4, 5}, Cof{4, 5}, Jui{1, 2, 4, 5}, Wat{1, 2, 4, 5}, Koo{1, 3, 4, 5}, Che{1, 2, 3, 4, 5}, OG{1, 2, 3, 4, 5}, LS{1, 2, 4, 5}, Par{2, 3, 4, 5}, Fox{1, 2, 3, 4, 5}, Hor{1, 2, 3, 4, 5}, Sna{1, 2, 3, 4, 5}, Dog{2, 3, 4, 5}, Zeb{1, 2, 3, 4, 5}	11. Color(i) = Ivo $\Leftrightarrow$ Color(i+1) = Gre (với $i \in \{1, 2, 3, 4\}$)
Nor{1}, Ukr{2, 4, 5}, Bri{3, 4, 5}, Spa{2, 3, 4, 5}, Jap{2, 3, 4, 5} Blu{2}, Red{3, 4, 5}, Gre{4, 5}, Ivo{3, 4}, Yel{1} Mil{3}, Tea{2, 4, 5}, Cof{4, 5}, Jui{1, 2, 4, 5}, Wat{1, 2, 4, 5},	16. All diff

Koo{1, 3, 4, 5}, Che{1, 2, 3, 4, 5}, OG{1, 2, 3, 4, 5}, LS{1, 2, 4, 5}, Par{2, 3, 4, 5}, Fox{1, 2, 3, 4, 5}, Hor{1, 2, 3, 4, 5}, Sna{1, 2, 3, 4, 5}, Dog{2, 3, 4, 5}, Zeb{1, 2, 3, 4, 5}	
Nor{1}, Ukr{2, 4, 5}, Bri{3, 4, 5}, Spa{2, 3, 4, 5}, Jap{2, 3, 4, 5} Blu{2}, Red{3, 4, 5}, Gre{4, 5}, Ivo{3, 4}, Yel{1} Mil{3}, Tea{2, 4, 5}, Cof{4, 5}, Jui{1, 2, 4, 5}, Wat{1, 2, 4, 5}, Koo{1}, Che{1, 2, 3, 4, 5}, OG{1, 2, 3, 4, 5}, LS{1, 2, 4, 5}, Par{2, 3, 4, 5}, Fox{1, 2, 3, 4, 5}, Hor{1, 2, 3, 4, 5}, Sna{1, 2, 3, 4, 5}, Dog{2, 3, 4, 5}, Zeb{1, 2, 3, 4, 5}	8. Tobacco(i) = Koo $\Leftrightarrow$ Color(i) = Yel
Nor{1}, Ukr{2, 4, 5}, Bri{3, 4, 5}, Spa{2, 3, 4, 5}, Jap{2, 3, 4, 5} Blu{2}, Red{3, 4, 5}, Gre{4, 5}, Ivo{3, 4}, Yel{1} Mil{3}, Tea{2, 4, 5}, Cof{4, 5}, Jui{1, 2, 4, 5}, Wat{1, 2, 4, 5}, Koo{1}, Che{1, 2, 3, 4, 5}, OG{1, 2, 3, 4, 5}, LS{1, 2, 4, 5}, Par{2, 3, 4, 5}, Fox{1, 2, 3, 4, 5}, Hor{2}, Sna{1, 2, 3, 4, 5}, Dog{2, 3, 4, 5}, Zeb{1, 2, 3, 4, 5}	13. Tobacco(i) = Koo $\Leftrightarrow$ [Pet(i-1) = Hor] $\vee$ [Pet(i+1) = Hor]
Nor{1}, Ukr{2, 4, 5}, Bri{3, 4, 5}, Spa{2, 3, 4, 5}, Jap{2, 3, 4, 5} Blu{2}, Red{3, 4, 5}, Gre{4, 5}, Ivo{3, 4}, Yel{1} Mil{3}, Tea{2, 4, 5}, Cof{4, 5}, Jui{1, 2, 4, 5}, Wat{1, 2, 4, 5},	16. All Diff

Koo{1}, Che{1, 2, 3, 4, 5}, OG{1, 2, 3, 4, 5}, LS{1, 2, 4, 5}, Par{2, 3, 4, 5}, Fox{1, 3, 4, 5}, Hor{2}, Sna{1, 3, 4, 5}, Dog{3, 4, 5}, Zeb{1, 3, 4, 5}	
Nor{1}, Ukr{2, 4, 5}, Bri{3, 4, 5}, Spa{3, 4, 5}, Jap{2, 3, 4, 5} Blu{2}, Red{3, 4, 5}, Gre{4, 5}, Ivo{3, 4}, Yel{1} Mil{3}, Tea{2, 4, 5}, Cof{4, 5}, Jui{1, 2, 4, 5}, Wat{1, 2, 4, 5}, Koo{1}, Che{1, 2, 3, 4, 5}, OG{1, 2, 3, 4, 5}, LS{1, 2, 4, 5}, Par{2, 3, 4, 5}, Fox{1, 3, 4, 5}, Hor{2}, Sna{1, 3, 4, 5}, Dog{3, 4, 5}, Zeb{1, 3, 4, 5}	4. Nationality(i) = Spa $\Leftrightarrow$ Pet(i) = Dog
Nor{1}, Ukr{2, 4, 5}, Bri{3, 4, 5}, Spa{3, 4, 5}, Jap{2, 3, 4, 5} Blu{2}, Red{3, 4, 5}, Gre{4, 5}, Ivo{3, 4}, Yel{1} Mil{3}, Tea{2, 4, 5}, Cof{4, 5}, Jui{1, 2, 4, 5}, Wat{1, 2, 4, 5}, Koo{1}, Che{2, 3, 4, 5}, OG{3, 4, 5}, LS{2, 4, 5}, Par{2, 3, 4, 5}, Fox{1, 3, 4, 5}, Hor{2}, Sna{1, 3, 4, 5}, Dog{3, 4, 5}, Zeb{1, 3, 4, 5}	Tobacco(i) = OG $\Leftrightarrow$ Pet(i) = Sna 16. All diff
Nor{1}, Ukr{2, 4, 5}, Bri{3, 4, 5}, Spa{3, 4, 5}, Jap{2, 3, 4, 5} Blu{2}, Red{3, 4, 5}, Gre{4, 5}, Ivo{3, 4}, Yel{1} Mil{3}, Tea{2, 4, 5}, Cof{4, 5}, Jui{1, 2, 4, 5}, Wat{1, 2, 4, 5},	7. Tobacco(i) = OG $\Leftrightarrow$ Pet(i) = Sna

Koo{1}, Che{2, 3, 4, 5}, OG{3, 4, 5}, LS{2, 4, 5}, Par{2, 3, 4, 5}, Fox{1, 3, 4, 5}, Hor{2}, Sna{3, 4, 5}, Dog{3, 4, 5}, Zeb{1, 3, 4, 5}	
Nor{1}, Ukr{2, 4, 5}, Bri{3, 4, 5}, Spa{3, 4, 5}, Jap{2, 3, 4, 5} Blu{2}, Red{3, 4, 5}, Gre{4, 5}, Ivo{3, 4}, Yel{1} Mil{3}, Tea{2, 4, 5}, Cof{4, 5}, Jui{2, 4, 5}, Wat{1, 2, 4, 5}, Koo{1}, Che{2, 3, 4, 5}, OG{3, 4, 5}, LS{2, 4, 5}, Par{2, 3, 4, 5}, Fox{1, 3, 4, 5}, Hor{2}, Sna{3, 4, 5}, Dog{3, 4, 5}, Zeb{1, 3, 4, 5}	9. Tobacco(i) = LS $\Leftrightarrow$ Drink(i) = Jui
Nor{1}, Ukr{2, 4, 5}, Bri{3, 4, 5}, Spa{3, 4, 5}, Jap{2, 3, 4, 5} Blu{2}, Red{3, 4, 5}, Gre{4, 5}, Ivo{3, 4}, Yel{1} Mil{3}, Tea{2, 4, 5}, Cof{4, 5}, Jui{2, 4, 5}, Wat{1} Koo{1}, Che{2, 3, 4, 5}, OG{3, 4, 5}, LS{2, 4, 5}, Par{2, 3, 4, 5}, Fox{1, 3, 4, 5}, Hor{2}, Sna{3, 4, 5}, Dog{3, 4, 5}, Zeb{1, 3, 4, 5}	16. ALL diff
Nor{1}, Ukr{2, 4, 5}, Bri{3, 4, 5}, Spa{3, 4, 5}, Jap{2, 3, 4, 5} Blu{2}, Red{3, 4, 5}, Gre{4, 5}, Ivo{3, 4}, Yel{1} Mil{3}, Tea{2, 4, 5}, Cof{4, 5}, Jui{2, 4, 5}, Wat{1}	15. Tobacco(i) = Che $\Leftrightarrow$ [Drink(i-1) = Wat] $\vee$ [Drink(i+1) = Wat]

Koo{1}, Che{2}, OG{3, 4, 5}, LS{2, 4, 5}, Par{2, 3, 4, 5}, Fox{1, 3, 4, 5}, Hor{2}, Sna{3, 4, 5}, Dog{3, 4, 5}, Zeb{1, 3, 4, 5}	
Nor{1}, Ukr{2, 4, 5}, Bri{3, 4, 5}, Spa{3, 4, 5}, Jap{2, 3, 4, 5} Blu{2}, Red{3, 4, 5}, Gre{4, 5}, Ivo{3, 4}, Yel{1} Mil{3}, Tea{2, 4, 5}, Cof{4, 5}, Jui{2, 4, 5}, Wat{1} Koo{1}, Che{2}, OG{3, 4, 5}, LS{4, 5}, Par{3, 4, 5}, Fox{1, 3, 4, 5}, Hor{2}, Sna{3, 4, 5}, Dog{3, 4, 5}, Zeb{1, 3, 4, 5}	16. ALL diff
Nor{1}, Ukr{2, 4, 5}, Bri{3, 4, 5}, Spa{3, 4, 5}, Jap{2, 3, 4, 5} Blu{2}, Red{3, 4, 5}, Gre{4, 5}, Ivo{3, 4}, Yel{1} Mil{3}, Tea{2, 4, 5}, Cof{4, 5}, Jui{2, 4, 5}, Wat{1} Koo{1}, Che{2}, OG{3, 4, 5}, LS{4, 5}, Par{3, 4, 5}, Fox{1, 3}, Hor{2}, Sna{3, 4, 5}, Dog{3, 4, 5}, Zeb{1, 3, 4, 5}	12. Tobacco(i) = Che $\Leftrightarrow$ [Pet(i-1) = Fox] $\vee$ [Pet(i+1) = Fox]
Nor{1}, Ukr{2, 4, 5}, Bri{3, 4, 5}, Spa{3, 4, 5}, Jap{2, 3, 4, 5} Blu{2}, Red{3, 4, 5}, Gre{4, 5}, Ivo{3, 4}, Yel{1} Mil{3}, Tea{2, 4, 5}, Cof{4, 5}, Jui{4, 5}, Wat{1}	9. Tobacco(i) = LS $\Leftrightarrow$ Drink(i) = Jui

Koo{1}, Che{2}, OG{3, 4, 5}, LS{4, 5}, Par{3, 4, 5}, Fox{1, 3}, Hor{2}, Sna{3, 4, 5}, Dog{3, 4, 5}, Zeb{1, 3, 4, 5}	
Nor{1}, Ukr{2, 4, 5}, Bri{3, 4, 5}, Spa{3, 4, 5}, Jap{3, 4, 5} Blu{2}, Red{3, 4, 5}, Gre{4, 5}, Ivo{3, 4}, Yel{1} Mil{3}, Tea{2, 4, 5}, Cof{4, 5}, Jui{4, 5}, Wat{1} Koo{1}, Che{2}, OG{3, 4, 5}, LS{4, 5}, Par{3, 4, 5}, Fox{1, 3}, Hor{2}, Sna{3, 4, 5}, Dog{3, 4, 5}, Zeb{1, 3, 4, 5}	10. Nationality(i) = Jap ⇔ Tobacco(i) = Par
Nor{1}, Ukr{2}, Bri{3, 4, 5}, Spa{3, 4, 5}, Jap{3, 4, 5} Blu{2}, Red{3, 4, 5}, Gre{4, 5}, Ivo{3, 4}, Yel{1} Mil{3}, Tea{2, 4, 5}, Cof{4, 5}, Jui{4, 5}, Wat{1} Koo{1}, Che{2}, OG{3, 4, 5}, LS{4, 5}, Par{3, 4, 5}, Fox{1, 3}, Hor{2}, Sna{3, 4, 5}, Dog{3, 4, 5}, Zeb{1, 3, 4, 5}	16. All diff
Nor{1}, Ukr{2}, Bri{3, 4, 5}, Spa{3, 4, 5}, Jap{3, 4, 5} Blu{2}, Red{3, 4, 5}, Gre{4, 5}, Ivo{3, 4}, Yel{1} Mil{3}, Tea{2}, Cof{4, 5}, Jui{4, 5}, Wat{1}	6. Nationality(i) = Ukr ⇔ Drink(i) = Tea

Koo{1}, Che{2}, OG{3, 4, 5}, LS{4, 5}, Par{3, 4, 5}, Fox{1, 3}, Hor{2}, Sna{3, 4, 5}, Dog{3, 4, 5}, Zeb{1, 3, 4, 5}	
Nor{1}, Ukr{2}, Bri{3, 4, 5}, Spa{3, 4, 5}, Jap{3, 5} Blu{2}, Red{3, 4, 5}, Gre{5}, Ivo{3, 4}, Yel{1} Mil{3}, Tea{2}, Cof{5}, Jui{4}, Wat{1} Koo{1}, Che{2}, OG{3, 4, 5}, LS{4}, Par{3, 5}, Fox{1, 3}, Hor{2}, Sna{3, 4, 5}, Dog{3, 4, 5}, Zeb{1, 3, 4, 5}	Backtrack to Jui = 4
Nor{1}, Ukr{2}, Bri{3, 4, 5}, Spa{3, 4, 5}, Jap{3, 5} Blu{2}, Red{3, 4, 5}, Gre{5}, Ivo{4}, Yel{1} Mil{3}, Tea{2}, Cof{5}, Jui{4}, Wat{1} Koo{1}, Che{2}, OG{3, 4, 5}, LS{4}, Par{3, 5}, Fox{1, 3}, Hor{2}, Sna{3, 4, 5}, Dog{3, 4, 5}, Zeb{1, 3, 4, 5}	11. Color(i) = Ivo $\Leftrightarrow$ Color(i+1) = Gre (với $i \in \{1, 2, 3, 4\}$)
Nor{1}, Ukr{2}, Bri{3, 4, 5}, Spa{3, 4, 5}, Jap{3, 5} Blu{2}, Red{3}, Gre{5}, Ivo{4}, Yel{1} Mil{3}, Tea{2}, Cof{5}, Jui{4}, Wat{1} Koo{1}, Che{2}, OG{3, 4, 5}, LS{4}, Par{3, 5}, Fox{1, 3}, Hor{2}, Sna{3, 4, 5}, Dog{3, 4, 5}, Zeb{1, 3, 4, 5}	ALL Diff

Nor{1}, Ukr{2}, Bri{3}, Spa{3, 4, 5}, Jap{3, 5} Blu{2}, Red{3}, Gre{5}, Ivo{4}, Yel{1} Mil{3}, Tea{2}, Cof{5}, Jui{4}, Wat{1} Koo{1}, Che{2}, OG{3, 4, 5}, LS{4}, Par{3, 5}, Fox{1, 3}, Hor{2}, Sna{3, 4, 5}, Dog{3, 4, 5}, Zeb{1, 3, 4, 5}	3. Nationality(i) = Bri ⇔ Color(i) = Red
Nor{1}, Ukr{2}, Bri{3}, Spa{4}, Jap{5} Blu{2}, Red{3}, Gre{5}, Ivo{4}, Yel{1} Mil{3}, Tea{2}, Cof{5}, Jui{4}, Wat{1} Koo{1}, Che{2}, OG{3, 4, 5}, LS{4}, Par{3, 5}, Fox{1, 3}, Hor{2}, Sna{3, 4, 5}, Dog{3, 4, 5}, Zeb{1, 3, 4, 5}	16. All diff
Nor{1}, Ukr{2}, Bri{3}, Spa{4}, Jap{5} Blu{2}, Red{3}, Gre{5}, Ivo{4}, Yel{1} Mil{3}, Tea{2}, Cof{5}, Jui{4}, Wat{1} Koo{1}, Che{2}, OG{3, 4, 5}, LS{4}, Par{3, 5}, Fox{1, 3}, Hor{2}, Sna{3, 4, 5}, Dog{4}, Zeb{1, 3, 4, 5}	4. Nationality(i) = Spa ⇔ Pet(i) = Dog
Nor{1}, Ukr{2}, Bri{3}, Spa{4}, Jap{5} Blu{2}, Red{3}, Gre{5}, Ivo{4}, Yel{1} Mil{3}, Tea{2}, Cof{5}, Jui{4}, Wat{1} Koo{1}, Che{2}, OG{3, 4, 5}, LS{4}, Par{3, 5}, Par{5},	10. Nationality(i) = Jap ⇔ Tobacco(i) = Par

Fox{1, 3}, Hor{2}, Sna{3, 4, 5}, Dog{4}, Zeb{1, 3, 4, 5}	
Nor{1}, Ukr{2}, Bri{3}, Spa{4}, Jap{5} Blu{2}, Red{3}, Gre{5}, Ivo{4}, Yel{1} Mil{3}, Tea{2}, Cof{5}, Jui{4}, Wat{1} Koo{1}, Che{2}, OG{3} , LS{4}, Par{5}, Fox{1, 3}, Hor{2}, Sna{3, 4, 5}, Dog{4}, Zeb{1, 3, 4, 5}	16. All diff
Nor{1}, Ukr{2}, Bri{3}, Spa{4}, Jap{5} Blu{2}, Red{3}, Gre{5}, Ivo{4}, Yel{1} Mil{3}, Tea{2}, Cof{5}, Jui{4}, Wat{1} Koo{1}, Che{2}, OG{3}, LS{4}, Par{5}, Fox{1, 3}, Hor{2}, Sna{3} , Dog{4}, Zeb{1, 3, 4, 5}	7. Tobacco(i) = OG ⇔ Pet(i) = Sna
Nor{1}, Ukr{2}, Bri{3}, Spa{4}, Jap{5} Blu{2}, Red{3}, Gre{5}, Ivo{4}, Yel{1} Mil{3}, Tea{2}, Cof{5}, Jui{4}, Wat{1} Koo{1}, Che{2}, OG{3}, LS{4}, Par{5}, Fox{1} , Hor{2}, Sna{3}, Dog{4}, Zeb{5}	16. All diff

H(5) owns the zebra

H(1) drinks water